

DTFT 推导.

$$X(f) = \mathcal{F}[x(t) \cdot h(t)] = \int_{-\infty}^{\infty} x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t-nT) \cdot e^{-j2\pi f t} dt$$

$$= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} x(t) \cdot \delta(t-nT) e^{-j2\pi f t} dt$$

$$= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} x(nT) \delta(t-nT) \cdot e^{-j2\pi f nT} dt$$

$$= \sum_{n=-\infty}^{\infty} x(nT) e^{-j2\pi f nT} \int_{-\infty}^{\infty} \delta(t-nT) dt$$

$$= \sum_{n=-\infty}^{\infty} x(nT) \cdot e^{-j2\pi f nT}$$